

Ford-Fulkerson Algorithm **(Max-Flow Algorithm)**

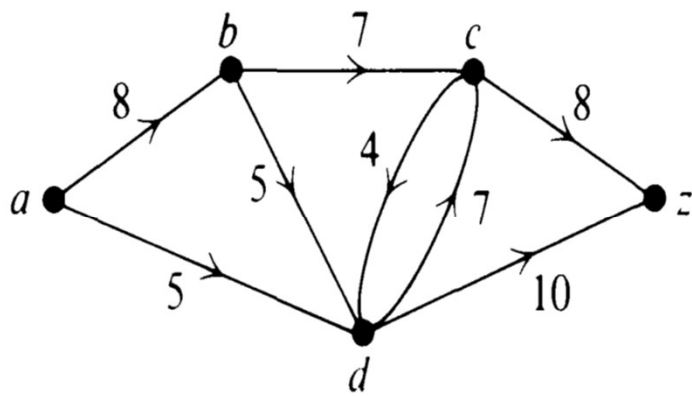
TRANSPORT NETWORK

A weighted directed graph is said to be a *transport network* if the following conditions are satisfied:

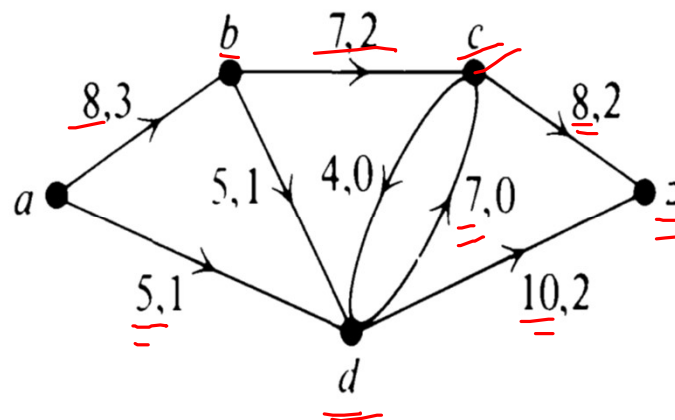
1. It is connected and contains no loops.
2. There is one and only one vertex in the graph that has no incoming edge.
3. There is one and only one vertex in the graph that has no outgoing edge.
4. The weight of each edge is a nonnegative real number.

TERMINOLOGIES

In a transport network, the vertex that has no incoming edge is called the *source* and is denoted by a ; the vertex that has no outgoing edge is called the *sink* and is denoted by z . The weight of an edge is called the *capacity* of the edge. The capacity of the edge (i, j) is denoted by $w(i, j)$.



(a)



(b)

FLOW IN TRANSPORT NETWORK

A *flow* in a transport network, ϕ , is an assignment of a nonnegative number $\phi(i, j)$ to each edge (i, j) such that the following conditions are satisfied:

1. $\phi(i, j) \leq w(i, j)$ for each edge (i, j) .
2. $\sum_{\text{all } i} \phi(i, j) = \sum_{\text{all } k} \phi(j, k)$ for each vertex j except the source a and the sink z .†



In terms of the transportation of material, $\phi(i, j)$ is the amount of material to be shipped through the route (i, j) . Condition 1 means that the amount of material to be shipped through a route cannot exceed the capacity of the route.

Condition 2 means that, except at the source and at the sink, the amount of material flowing into a vertex must equal the amount of material flowing out of the vertex. For example, Fig. 6.22b shows a flow in the transport network in Fig. 6.22a. The first number associated with an edge is the capacity of the edge, and the second number associated with an edge is the flow in the edge. The quantity $\sum_{\text{all } i} \phi(a, i)$ is said to be the *value of the flow* ϕ and is denoted by ϕ_v . Intuitively, it is clear that

$$\phi_v = \sum_{\text{all } i} \phi(a, i) = \sum_{\text{all } k} \phi(k, z)$$

LABELLING PROCEDURE FOR FINDING MAXIMUM FLOW IN THE NETWORK

At first, the source vertex a is labeled $(-\infty)$

All vertices that are adjacent from a are scanned.

A vertex b that is adjacent from a is labeled $(a, \Delta b)$ where $\Delta b = w(a, b) - \phi(a, b)$, if $w(a, b) > \phi(a, b)$, it is not labeled if $w(a, b) = \phi(a, b)$

After all the vertices that are adjacent to or from the labeled vertices are scanned

Let b is labeled vertex and q be the adjacent vertex from b

The vertex q is labeled as $(b^+, \Delta q)$ where Δq is smaller of two quantities Δb and $[w(b,q) - \phi(b,q)]$,
if $w(b,q) > \phi(b,q)$ it is not labeled if $w(b,q) = \phi(b,q)$

The vertex q is labeled as $(b^-, \Delta q)$ where Δq is smaller of two quantities Δb and $\phi(q,b)$, if $\phi(q,b) > 0$

The vertex q is not labeled if $\phi(q,b) = 0$

Such a labelling is not necessarily unique.

The vertex q might be adjacent to more than one vertex.

